

**Random forest regression using autonomous in situ ocean observations:
Inferring small-scale nutrient variability during a Southern Ocean field
experiment**

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15 ABSTRACT: Small-scale ocean processes are known to modulate nutrient availability in the
16 upper ocean, yet nutrient variability at scales of $O(1)$ day and $O(1)$ km has remained challenging to
17 characterize from observations. Here, we show that small-scale nutrient information can be inferred
18 from the 2019 Southern Ocean Glider Observations of Submesoscales (SOGOS) field experiment
19 using random forest regression (RFR). In this experiment, a Biogeochemical-Argo float with coarse-
20 resolution nutrient sensing capabilities was co-deployed with two rapid-sampling Seagliders to
21 autonomously observe a turbulent region by the Southwest Indian Ridge over ~ 80 days in austral
22 winter. Since the Seagliders did not measure nitrate directly, we used RFR to estimate nitrate from
23 temperature, salinity, pressure, and oxygen data. The RFR was locally trained using nearby ship-
24 based and Biogeochemical-Argo float observations to capture region-specific relationships between
25 these variables. When tested on nitrate data from the SOGOS float, our RFR reproduced these
26 measurements with high fidelity (95% of absolute errors $\leq 0.72 \mu\text{mol kg}^{-1}$) and outperformed
27 existing, widely-used global nutrient estimation routines trained primarily on ship-based data.
28 Applying the RFR to the Seaglider observations then generated novel nitrate distributions extending
29 to 1000 m with horizontal resolution of ~ 1.5 km and 2–3 hours compared to ~ 75 km and 5 days for
30 the SOGOS float. Our analysis of high-resolution machine learning estimates of Seaglider nitrate,
31 coupled with Seaglider observations of other properties, suggests that vigorous small-scale stirring
32 enhances upper ocean nutrient availability in this region and highlights the importance of oceanic
33 turbulence in mediating biological productivity.

34 SIGNIFICANCE STATEMENT: Rapid fluctuations in ocean nutrient content can be intense but
35 difficult to observe directly. Here, we infer high-frequency nutrient information from a Southern
36 Ocean field experiment by using regression to leverage multiple types of instruments. We utilize
37 local nitrate measurements from two observing platforms (ship and autonomous Argo floats)
38 to train a random forest regression that will estimate unknown nitrate concentrations along the
39 tracks of different autonomous instruments with significantly faster sampling rates (Seagliders).
40 Our targeted application produces novel nitrate estimates at high enough resolution to observe
41 strong, small-scale nutrient stirring as the Seagliders traverse a turbulent oceanic region. Increased
42 observational coverage of the ocean, combined with machine learning approaches, can thus enable
43 ocean analyses at increasingly fine scales.

44 1. Introduction

45 Nutrient availability exerts a significant control on oceanic primary productivity. The distribution
46 of nutrients in the subsurface ocean is mediated in part by physical processes that span a diverse
47 range of temporal and spatial scales. At the largest basin-wide scales, nutrient concentrations
48 are broadly determined by upwelling and downwelling motions associated with the wind-driven
49 circulation. At the oceanic mesoscale (horizontal scales of $O(20\text{--}200\text{ km})$, evolving over weeks to
50 months), horizontal and vertical nutrient transport processes are associated with localized features
51 representing anomalies in the circulation, called eddies, that modify the density structure of the
52 upper ocean (Su et al. 2021; Patel et al. 2020; Levy and Martin 2013; Mahadevan and Archer 2000).
53 Widespread mesoscale eddies subject local nutrient distributions to competing processes that are
54 both depth-dependent (e.g. utilization by primary production and recycling by remineralization)
55 and density-dependent (transport along surfaces of constant density, or isopycnals) (Omand and
56 Mahadevan 2013). For nutrients whose low availability in the upper ocean limits photosynthetic
57 activity, eddy-associated transport can enable more primary production in the surface mixed layer
58 (McGillicuddy et al. 1998).

59 Over smaller distances and shorter time periods, nutrient distributions are also impacted by
60 motions at the oceanic submesoscale (scales of $O(1\text{ day})$ and $O(1\text{--}10\text{ km})$; Lévy et al. 2024,
61 2018; Mahadevan 2016; Lévy et al. 2012; Thomas et al. 2008; Mahadevan and Tandon 2006).
62 Submesoscale dynamics are associated with strong vertical velocities, typically localized to fronts

63 and filamentary structures, that cause vigorous exchange between the interior and surface ocean
64 (Taylor and Thompson 2023; Freilich and Mahadevan 2019; Thomas et al. 2008; Brannigan 2016).
65 These processes can exert a strong influence on local primary production and carbon export since
66 light and nutrient availability change rapidly with depth (Lévy et al. 2018; Klein and Lapeyre 2009;
67 Erickson and Thompson 2018). Nutrient transport at submesoscales is particularly important for
68 mediating biological processes since these small-scale motions occur on the same timescales as
69 phytoplankton growth (Mahadevan 2016). Stirring along strongly tilted isopycnals, particularly
70 at submesoscale fronts, can drive large nutrient fluxes depending on the variation in nutrient
71 concentrations with depth (Freilich and Mahadevan 2019).

72 In this study, we focus on observations from a turbulent subregion of the Southern Ocean with
73 high surface and subsurface variability. The Southern Ocean, defined as the oceanic area south of
74 $\sim 35^\circ$ S, plays a central role in the planetary climate system due to its distinctive physical circulation
75 and biogeochemical characteristics (Henley et al. 2020; Rintoul and Naveira Garabato 2013; Talley
76 2013). The physical dynamics of the Southern Ocean, which are dominated by the strong flow
77 of the Antarctic Circumpolar Current as well as widespread mesoscale eddies and submesoscale
78 fronts, significantly impact the carbon cycle both regionally and globally (Gray 2024). Southern
79 Ocean model simulations have suggested that submesoscale stirring promotes exchange between
80 the nutrient-depleted surface ocean and deeper waters that are enriched in remineralized nutrients
81 (Uchida et al. 2019; Balwada et al. 2018). One experiment found that increasing a Southern Ocean
82 biogeochemical model's resolution from 100 km to 2 km nearly doubled primary production
83 due to nutrient injection into the mixed layer (Uchida et al. 2020). Other numerical simulations
84 indicate that where currents interact strongly with seafloor topography, submesoscale upwelling
85 may be the predominant mechanism by which dissolved iron reaches the upper Southern Ocean
86 (Rosso et al. 2016). These turbulent contributions to the nutrient supply have global effects
87 since the Southern Ocean is a central site for the formation of water masses with widespread
88 geographic distribution (Talley 2013). The challenges of sampling at the resolution required to
89 resolve submesoscale variability have previously limited our ability to characterize short-lived,
90 small-scale biogeochemical interactions using observations.

91 New observing technologies, including a variety of autonomous platforms, have emerged to
92 address this knowledge gap. Of these, profiling floats are instruments that drift freely with the

horizontal ocean currents and move vertically in the water column by controlling their buoyancy. The floats measure “profiles” of temperature and salinity as a function of pressure that typically extend to at least ~2000 m depth. Deployment of thousands of these instruments over the past two decades through the International Argo Programme has revolutionized our ability to observe and understand the global ocean (e.g. Johnson et al. 2022; Wong et al. 2020; Swart et al. 2023). The use of Biogeochemical-Argo (BGC-Argo) floats, which are equipped with additional biogeochemically-relevant sensors, has accelerated rapidly in recent years, and this type of float now constitutes an important component of the global Argo array (Sarmiento et al. 2023; Claustre et al. 2020; Roemmich et al. 2019). The Argo floats’ characteristic 10-day sampling period makes them well-suited for observing processes at the mesoscale and larger.

Underwater gliders also collect depth-resolved measurements of the ocean interior. In contrast to profiling floats, gliders can move both vertically and horizontally through the water, which enables substantial navigational control via remote piloting. Given this maneuverability, ocean gliders are capable of capturing small-scale variability, and glider-based observations have already provided important insights into submesoscale physical processes (e.g. Balwada et al. 2024; Swart et al. 2023; du Plessis M. D. et al. 2022; Erickson and Thompson 2018; Rudnick 2016; Thompson et al. 2017, 2016; Erickson et al. 2016). Though some gliders have been used to observe biogeochemical properties (e.g. Thompson et al. 2024; Possenti et al. 2021; Vincent et al. 2018; Birchill et al. 2021), most to date have not been deployed with a full suite of sensors due to challenges with power demand, temporal responsiveness, and measurement drift (Chai et al. 2020). While multiple types of gliders exist, the field campaign explored in this study employs two Seagliders that sample to a depth of 1000 m.

Here, we develop a random forest regression (RFR) to infer unobserved nutrient information from the Southern Ocean Glider Observations of Submesoscales (SOGOS) field experiment in 2019, during which two Seagliders and a BGC-Argo float sampled a highly energetic region by the Southwest Indian Ridge. This area is characterized by some of the highest values of mesoscale eddy kinetic energy (EKE) in the global ocean, and the presence of vigorous mesoscale eddies suggests that this region will feature strong nitrate stirring events. Since the SOGOS Seagliders capture small-scale variability in other ocean properties but do not measure nitrate directly, our RFR aims to reconstruct high-resolution nitrate distributions using the Seagliders’ available observations

123 of pressure, temperature, salinity, and oxygen. RFR has been successfully applied to diverse
124 oceanographic cases (e.g. Lyman and Johnson 2023; Sharp et al. 2023, 2022; Callens et al. 2020;
125 Tong et al. 2019; Gregor et al. 2019, 2017), including for Southern Ocean oxygen prediction using
126 BGC-Argo data (Giglio et al. 2018). The SOGOS experiment presents a natural opportunity for
127 supervised regression because nitrate observations from the co-sampling BGC-Argo float can be
128 used to assess model performance within the spatiotemporal context of interest.

129 To optimize the RFR for application to the SOGOS Seagliders, we design our regression model
130 to consider training data centered around this particular field campaign. Given the highly non-
131 stationary nature of ocean properties, locally trained regressions have been shown to be effective
132 for estimation of physical and biogeochemical fields (e.g. Carter et al. 2016, 2021; Kuusela and
133 Stein 2018). The spatiotemporal specificity of ocean nutrient dynamics is one reason that global
134 biogeochemical estimation routines, trained on ship-based data that are strongly concentrated in
135 the Northern Hemisphere, struggle to represent the Southern Ocean (Heimdal et al. 2024; Gloege
136 et al. 2021). Our RFR takes advantage of training data from BGC-Argo floats, which fill signif-
137 icant observing gaps in this region, to learn the locally relevant mapping of other ocean features
138 to nitrate. This float-based approach to reconstructing Seaglider nutrient distributions assumes
139 that the relationships captured by the BGC-Argo floats represent the dynamics of interest; this
140 assumption is reasonable since both autonomous platforms return instantaneous observations, as
141 opposed to averages over the time period between surfacing (i.e. 10 days or 4–6 hours). The
142 floats’ measurements are capturing submesoscale variability, though they are not able to resolve
143 submesoscale dynamics in isolation (Llort et al. 2018). Our RFR, trained to estimate vertical
144 profiles using instantaneous float measurements, is therefore fully applicable to the glider measure-
145 ments regardless of diving frequency and horizontal spacing. We utilize the RFR to derive novel
146 Seaglider-based nitrate distributions, depth-resolved to 1000 m every 2–3 hours, that we then use
147 to investigate small-scale, biogeochemical-physical interactions during the SOGOS deployment.

148 This study is presented in four sections, the first of which describes the observational data
149 (Section 2). The methods are detailed in Section 3, with explanations of the RFR methodology
150 in Section 3a, of model evaluation using synthetic observations in Section 3b, and of time series
151 and wavelet analyses used to characterize the RFR nitrate in Section 3c–d. In Section 4a, we
152 train, validate, and test the RFR before applying it to the SOGOS Seaglider inputs in Section 4b

to generate high-resolution nitrate distributions. Section 4b.1 highlights relationships between physical and biological time series in the upper ocean, while Section 4b.2 explores timescales of nutrient variability. We conclude in Section 5 by commenting on the utility of multi-platform deployments, as well as RFR applicability to other oceanographic observing programs.

2. Data

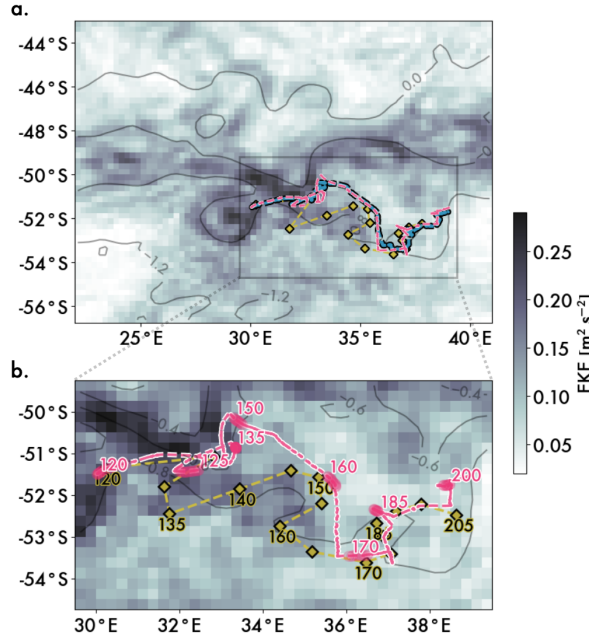


FIG. 1: (a) Trajectories for Seagliders SG659 (blue) and SG660 (magenta) and SOGOS float profile locations (yellow diamonds). Gray box denotes the area in panel (b), showing an expanded view of the float (yellow) and SG660 (magenta) as they traverse from the high EKE region (yeardays 120–150) to the low EKE region (yeardays 170–200). Magenta (yellow) text indicates yearday of the highlighted SG660 (float) observations. Background shading represents eddy kinetic energy (EKE, $\text{m}^2 \text{s}^{-2}$) averaged over the Seaglider deployment; gray contours represent averaged satellite sea surface height (m).

During the 2019 SOGOS experiment, two Seagliders (SG659 and SG660) and a BGC-Argo float (World Meteorological Organization (WMO) ID 5906030; hereafter, the SOGOS float) co-sampled an especially turbulent region of the Southern Ocean by the Southwest Indian Ridge (SWIR), a known area of enhanced eddy kinetic energy (EKE) (Balwada et al. 2024; Yung et al. 2022). After being deployed from the I06 Global Ship-based Hydrographic Investigations Program (GO-SHIP) cruise around 51.5° S and 30° E, the three SOGOS instruments were carried eastward by the Antarctic Circumpolar Current. The Seagliders sampled for 86 days (May 1, 2019–July 25, 2019;

165 austral fall to winter), covering a region spanning approximately 30–40° E and 50–54° S (Figure
166 1a). Further details on the deployment and trajectories of the Seagliders are given in Dove et al.
167 2021.

168 Seagliders take rapid measurements as they ascend and descend through the water column in
169 V-shaped “dives” and have been powerful observing tools for targeted process studies typically
170 lasting weeks–months, including extensively in the Southern Ocean (Swart et al. 2023). The two
171 SOGOS Seagliders were outfitted with temperature (T), salinity (S), and pressure (p) sensors on
172 an unpumped CTD (Conductivity-Temperature-Depth instrument; CT-Sail) and an oxygen (O₂)
173 optode (Aanderaa 4330 standard foil). Chlorophyll fluorescence and optical backscatter data at
174 470 nm and 700 nm were also collected (WETLabs ECO puck). SG659 and SG660 completed 456
175 and 502 V-shaped dives, respectively, to ~1000 m depth and surfaced every 4–6 hours (profiles
176 every 2–3 hours). Quality control of the Seaglider data is described in the Appendix. In total there
177 were 947,968 and 862,039 quality-controlled point observations with ~1 m vertical resolution from
178 SG659 and SG660, respectively.

179 Unlike Seagliders, BGC-Argo floats take measurements only during their ascent and return
180 vertical profiles of the ocean interior typically every 10 days. Standard BGC-Argo floats conserve
181 battery life by drifting with interior currents for ~9 days between profiles in order to provide
182 longer term coverage (average lifetimes of ~6 years) across the global ocean. All BGC-Argo floats
183 used in this study were equipped with sensors for T, S, p, O₂ as well as nitrate, fluorescence,
184 and backscatter. The T, S, and p data from the float CTD, which are continuously sampled at 1
185 Hz, are reported as averages over 2 dbar bins. In contrast, the biogeochemical data for the BGC-
186 Argo floats employed here are not subject to depth-binning since these sensors took instantaneous
187 spot samples (Riser et al. 2018; Wong et al. 2020). For these floats, the vertical sampling of
188 biogeochemical measurements ranged from every 5 dbar (≤ 100 dbar) to every 50 dbar (400–1000
189 dbar). During the SOGOS experiment, the SOGOS float sampled with 5-day profiling frequency,
190 returning 16 profiles. Observations from nine other BGC-Argo floats (referred to as “standard”
191 BGC-Argo floats; Figure 2a) were used for RFR training because these floats provided coverage of
192 the Antarctic Southern Zone bounded by the Polar Front and Sea Ice Edge (annual mean positions;
193 Sauvé et al. 2023) over 2017–2021. The float data, collected at the typical 10-day profiling
194 frequency, span two years before and after the SOGOS Seaglider deployment (May 8, 2017–July

11, 2021; Figure 2b). We limited the training and validation data to Antarctic Southern Zone float profiles between 5° E and 65° E, which were determined to be sufficiently close in space and time to the intended application (sensitivity analysis to size of spatiotemporal window in Supplemental Material; Figure S6). After quality control (Appendix), there were 22,558 standard BGC-Argo float and 3,644 SOGOS float point observations.

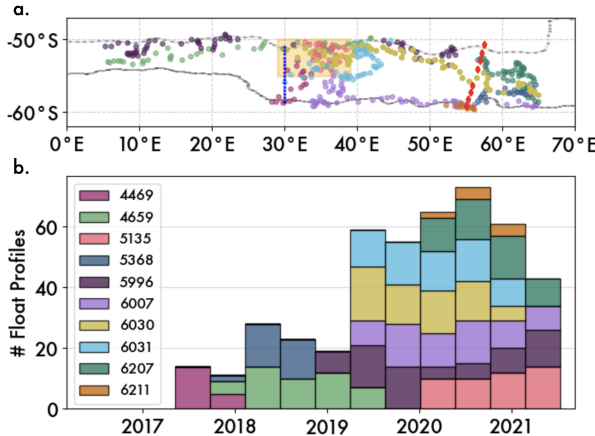


FIG. 2: (a) Location of standard BGC-Argo float profiles used for training, SOGOS float profiles used for testing (WMO ID: 5906030), GO-SHIP I06 station locations used for training (blue squares), and I07 station locations used for testing (red diamonds). Gray lines denote the 2019 mean positions of the Polar Front and Sea Ice Edge (Sauvé et al. 2023). (b) Time coverage by the BGC-Argo floats (May 2017 to July 2021). “590” truncated from float WMO labels in legend.

Ship data from 16 stations along the GO-SHIP I06 2019 transect (246 point observations) and 8 stations from the I07 2019 transect (96 point observations) provided T and S measurements as well as discrete nitrate and O₂ concentrations from bottle data. The GO-SHIP bottle data were considered the highest-quality observations and used for calibration of other platforms. Observations from the I06 cruise, from which the SOGOS float and Seagliders were deployed, were included in training, while all observations from I07 were withheld for independent testing.

Data from all platforms were limited to the upper 1000 m and processed before inclusion in the RFR (full details in Appendix). Time was reported as days elapsed since January 1, 2019 (yearday). Latitude and longitude values were used in the RFR without transformation given the small size of the SOGOS study region. Conservative temperature (Θ), absolute salinity (S_A), potential density (σ_0), spice (τ), Brunt-Väisälä buoyancy frequency (N^2), and oxygen saturation (O_{2,sat}) were calculated using the Thermodynamic Equation of Seawater 2010 (TEOS-10) Python toolbox (IOC et al. 2010). Ocean spice (τ) is a calculated measure of temperature and salinity

213 variations that acts as a tracer of water masses in the interior ocean. Mixed layer depth (MLD) was
214 calculated as the shallowest depth at which the potential density was 0.05 kg m^{-3} greater than that
215 observed at 10 m, following Dove et al. (2021).

216 Surface EKE was characterized using the delayed-time multi-satellite gridded product ($1/4^\circ$ res-
217 olution) for sea level heights and derived variables produced and distributed by the Copernicus
218 Marine Environment Monitoring Service Data Unification and Altimeter Combination System
219 (CMEMS DUACS; <https://doi.org/10.48670/moi-00148>). The altimetry product provides zonal
220 and meridional geostrophic velocity anomalies (u'_g and v'_g , respectively) from which EKE was
221 calculated as $\frac{1}{2}\sqrt{u'^2_g + v'^2_g}$. Following the framework in Dove et al. 2021, we delineated a high
222 EKE region (yeardays 120–150; $\text{EKE} > 0.124 \text{ m}^2 \text{ s}^{-2}$) and a low EKE region (yeardays 170–200;
223 $\text{EKE} < 0.109 \text{ m}^2 \text{ s}^{-2}$) within the SOGOS study site.

224 In addition, the mixed layer analyses used satellite-based estimates of photosynthetically ac-
225 tive radiation (PAR) from Moderate Resolution Imaging Spectroradiometer (MODIS) satellite
226 data at 4 km resolution in 8-day composites (National Aeronautics and Space Administra-
227 tion/Goddard Space Flight Center (NASA/GSFC) Ocean Biology Processing Group; Dataset
228 ID: erdMH1par08day). To diagnose submesoscale features, we used the finite-size Lyapunov
229 exponent (FSLE) value-added Level-4 product from the delayed-time merged Global Ocean
230 Gridded Absolute Geostrophic Velocities product ($1/25^\circ$ resolution; backward-in-time) (LO-
231 CEAN/CLS/CTOH/CNES 2021). Large negative FSLE values correspond to high local strain
232 rates, so filaments of strongly negative FSLE indicate lines of intense stretching that constrain fluid
233 transport along submesoscale fronts and around coherent structures like mesoscale eddies (e.g.
234 d'Ovidio et al. 2004; Siegelman et al. 2020).

235 3. Methods

236 a. Random Forest Regression

237 Our regression approach assumes that there is a learnable relationship between hydrographic
238 data collected by the Seagliders and ocean nutrient concentrations, and that these same dynamics
239 are randomly sampled by BGC-Argo floats which measure nutrients directly. The objective of this
240 study was to train a supervised model to learn nitrate relationships specific to the Southern Ocean

region of interest, then apply the regression to a set of predictor variables (“features”) measured by the SOGOS Seagliders to estimate nitrate at submesoscale resolution.

From preliminary algorithm comparisons (benchmark comparison to multiple linear regression given in Supplemental Material; Figure S7), we focused on random forest regression (RFR), which offers an attractive balance of interpretability and complexity for capturing nonlinear relationships. In other oceanographic contexts, RFR has performed comparably to other machine learning methods like neural networks, gradient-boosting machines, and support vector regression while being less computationally expensive to train (e.g. Gregor et al. 2017, 2019; Sharp et al. 2023). RFR is an ensemble machine learning method that combines the results of n decision trees to estimate the value of a target variable (Breiman 2001), or nitrate in our case. RFR incorporates randomness by both (1) limiting the set of predictive features considered at each step in the construction of the trees, and (2) training each tree on a random sample of the total training data with replacement (bootstrapping). By decorrelating individual trees and averaging over a large set, RFR can reduce overfitting tendencies given spatiotemporally biased observations (Stock 2022).

We designed our RFR to minimize nitrate estimation error within the specific spatiotemporal context of the SOGOS experiment (Figure 3). Since the SOGOS float sampled close to the Seagliders by experimental design, we chose to withhold the SOGOS float observations (WMO ID 5906030) for testing (3,644 point observations). Our data for training and validation included nitrate measurements from GO-SHIP I06 and nine standard BGC-Argo floats (22,800 point observations in total). We considered different combinations of predictor features such as Θ , S_A , σ_0 , τ , N^2 , O_2 , $O_{2,sat}$, as well as coordinate features like latitude (lat), longitude (lon), time, and season (sz_1 and sz_2) (Figure S2). N^2 was log-transformed since its distribution is non-Gaussian. Seasonal information was encoded as two cyclical training variables, sz_1 and sz_2 , following Gregor et al. 2018:

$$\begin{aligned} sz_1 &= \cos(2\pi * yd/365), \\ sz_2 &= \sin(2\pi * yd/365), \end{aligned} \tag{1}$$

where yd refers to yearday (days elapsed) referenced to January 1, 2019. Each RFR was constructed with $n = 1000$ trees using its implementation as `RandomForestRegressor` in the Python package

scikit-learn; a random one-third subset of features were considered at each node, with a minimum
node sample size of 5.

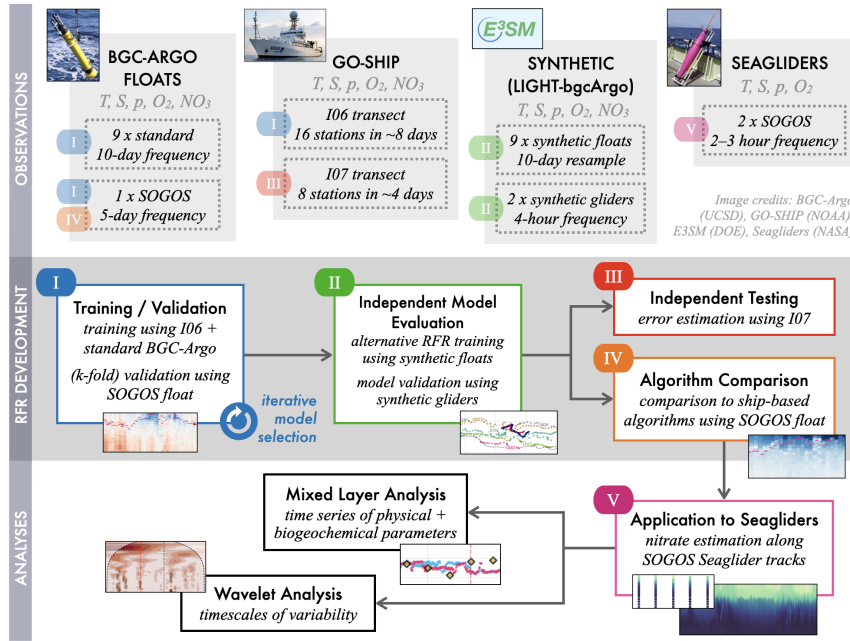


FIG. 3: Schematic of regression development using available GO-SHIP, BGC-Argo float, synthetic, and Seaglider observations. GO-SHIP I06 and BGC-Argo data were used for model training and validation, synthetic observations were used for independent model evaluation, GO-SHIP I07 and SOGOS float data were used for final model testing. The RFR was applied to the SOGOS Seaglider data to derive high-resolution nitrate maps to 1000 m depth.

The final RFR model parameters were determined based on three validation schemes (details in Supplemental Material). We screened an exhaustive set of feature lists using simple holdout validation, in which 20% of the available GO-SHIP and standard BGC-Argo data were withheld for error evaluation (Figure S3a). Since observations within a single vertical profile tend to be correlated, they were kept together during data splitting so that individual profiles are treated as independent units. To further evaluate performance for specified models, we applied k -fold validation, in which the RFR is iteratively trained on all but a random $1/k$ th sample (“fold”) of the data, e.g. trained using 90% of the training data for $k=10$ (Figure S3b). Errors computed on the withheld fold are combined over the k iterations to give error distributions (Kohavi 1995). We also considered potential spatiotemporal biases using spatial leave-one-out validation (Figure S3c), in which observations from one float are iteratively withheld for validation (Le Rest et al. 2014; Millard and Richardson 2015; Stock and Subramaniam 2022). After feature selection, we

281 tested the performance of our RFR on the independent SOGOS float data and compared it to that of
282 two existing, globally trained regression algorithms; an additional test was conducted using unseen
283 data from GO-SHIP I07 (93 point observations).

284 *b. Model Evaluation using Synthetic Observations*

285 To further evaluate the robustness of our model framework, we used data from synthetic bio-
286 geochemical floats (LIGHT-bgcArgo-1.0), which were recently developed to sample the Energy
287 Exascale Earth System Model version 2.1 (E3SMv2.1; Smith et al. 2025) directly during run time
288 and return instantaneous full-depth profiles at 4-hourly intervals (Nissen et al. 2024; Wolfram et al.
289 2015). Previous synthetic datasets were typically generated by sub-sampling the time-averaged
290 output of numerical models, often at monthly resolution, after run time. Significant temporal
291 smoothing of signals limited the ability of these products to represent high-frequency dynamics as
292 they would exist in the real ocean. The synthetic LIGHT-bgcArgo floats eliminated the uncertain-
293 ties introduced by offline sampling and better imitated the snapshot sampling of true autonomous
294 instruments.

295 Here, we used data retrieved instantaneously during a E3SMv2.1 model simulation run on a grid
296 with eddy-permitting resolution in the Southern Ocean; the synthetic float setup is described in
297 Nissen et al. 2024. Data from nine random synthetic floats whose trajectories passed through the
298 same spatiotemporal bounds of the real training data were sub-sampled at the typical BGC-Argo
299 10-day resolution at a randomized time of day. As before, the training data were restricted to the
300 Antarctic Southern Zone with 0.5° latitudinal buffer (approximate spatial bounds 5° E–65° E and
301 49° S–60° S; Figure 4a) and time period May 8, 2017–July 11, 2021. This synthetic float dataset
302 was then used to train an additional model, which we referred to as the synthetic RFR. A test dataset
303 was generated using two different synthetic floats (synSG-1, synSG-2) which sampled through the
304 SOGOS study region over a similar time of year as the Seagliders (April–July 2019). Unlike
305 the resampled training data, the test data were kept at the original 4-hourly resolution to mimic
306 the 2–3 hourly Seaglider profiling frequency. All synthetic datasets were restricted to pressures
307 ≤ 1000 dbar.

308 *c. Mixed Layer Characterization*

309 We applied the SOGOS RFR to both Seaglider datasets to produce new nitrate distributions
 310 at high spatiotemporal resolution. Using the RFR estimates and other measured properties, we
 311 characterized various time series in the mixed layer to explore physical-biogeochemical interactions
 312 over the SOGOS deployment. Along-track Seaglider EKE was calculated by interpolating satellite
 313 surface EKE to average profile locations. The absolute horizontal buoyancy gradient in the mixed
 314 layer ($|\nabla_h b|$) was estimated for each i th profile following $|\nabla_h b|_i = (|b_{i+1} - b_{i-1}|)(\Delta d)^{-1}$, where b_i is
 315 the mean mixed layer buoyancy for the i th profile and Δd is the horizontal distance between profiles
 316 at indices $i - 1$ and $i + 1$. Overall, $\nabla_h b$ underestimates the strength of fronts because the Seagliders
 317 do not always sample these features perpendicularly (Thompson et al. 2016). However, large
 318 outlying $\nabla_h b$ values can occur when the depth-averaged current causes the Seaglider separation
 319 Δd to be small due to the surfacing position being close to the dive position. We chose not to
 320 account for the impact of the depth-averaged current in this analysis.

321 We represented mixed layer mean nitrate ($\bar{N}_{ML}$, in $\mu\text{mol kg}^{-1}$) as integrated nitrate content in the
 322 mixed layer, normalized by mixed layer thickness (h_{ML} , in m):

$$\bar{N}_{ML} = \frac{1}{\rho_0} \frac{1}{h_{ML}} \int_{-h_{ML}}^0 [NO_3^-] \rho \, dz, \quad (2)$$

323 where $[NO_3^-](z)$ is the RFR-nitrate prediction in $\mu\text{mol kg}^{-1}$, $\rho(z)$ is in situ density in kg m^{-3} , and ρ_0
 324 (1027 kg m^{-3}) is an approximate seawater density near the surface for conversion into concentration.
 325 The integral was estimated using trapezoidal sums on the irregularly vertically spaced observations.
 326 We approximated the uncertainty ($\sigma_{\bar{N}}$) in each profile's $\bar{N}_{ML}$ as $\sigma_{\bar{N}}^2 = \sigma_{RFR}^2 \sum (h_{ML}^{-1} \Delta z_i)^2$ where
 327 σ_{RFR} is the RFR nitrate uncertainty and Δz_i is the spacing between measurements within the mixed
 328 layer (usually 1 m for the gliders). The RFR errors vary slightly with depth (Figure 6d), so σ_{RFR}
 329 for the mixed layer is approximated using the median absolute test errors from the SOGOS float,
 330 limited to the upper 200 m. This estimate does not consider correlation in the uncertainties with
 331 depth and is thus a lower bound. To provide a 95% confidence interval (CI) in $\bar{N}_{ML}$, we report
 332 1.95 times the mean of glider profile $\sigma_{\bar{N}}$ values.

333 We also analyzed the difference in nitrate concentration across the base of the mixed layer
 334 (ΔN_{ML}), defined as $\bar{N}_{ML}$ minus the mean nitrate concentration 20 m below the MLD ($\bar{N}_{ML+20}$;

averaged over 10–30 m below MLD). Uncertainty ($\sigma_{\Delta N}$) is estimated as $\sigma_{\Delta N}^2 = \sigma_N^2 + \sigma_{\overline{N}_{ML+20}}^2$, which is an upper bound because we have not accounted for correlations with depth; the 95% confidence interval is calculated as above. Horizontal variance of mixed layer nitrate (s_{H,NO_3}^2 , in $\mu\text{mol}^2\text{kg}^{-2}$) was calculated as the variance in $\overline{N}_{ML}$ when binned into 10-profile (~ 2 day) windows that overlap by 2 profiles (~ 10 hours). We used optical backscatter (scattering of light by suspended particles) as a proxy for particulate organic carbon and therefore biological production; the mixed layer mean optical backscatter at 470 nm ($\overline{bbp}_{470}$) was calculated analogously to Equation 2.

d. Wavelet Analyses

We explored important timescales of nitrate variability in the SOGOS RFR estimates using continuous wavelet transform (CWT). Time-localized wavelet analysis is useful for oceanographic data because of non-stationarity and persistence in the signals (Grinsted et al. 2004; Torrence and Compo 1998). Here, we used weighted-wavelet Z transform (WWZ; Foster 1996) to decompose the time series into dilated and shifted transformations of a complex Morlet wavelet (sinusoidal wave with Gaussian decay). Since Seagliders move in V-shaped dives (Section 2), their measurements along a given density surface (isopycnal) or depth are not associated with a regular sampling frequency; WWZ is well-suited for this application because it does not require interpolation of irregularly spaced input data (Torrence and Compo 1998). We implemented WWZ using the Python package Pyleoclim designed for use on climate time series (Khider et al. 2022). Significance testing was done by comparison to CWTs of 1000 iterations of the theoretical red noise, first order autoregressive process AR(1) benchmark, which is appropriate for ocean processes with temporal correlations (Torrence and Compo 1998).

4. Results

a. Random Forest Regression

Our multi-step RFR model validation (Figure S4) using GO-SHIP and standard BGC-Argo float observations led to the choice of 9 features, Θ , S_A , p , O_2 , latitude, longitude, time, sz_1 , and sz_2 . The results of this process suggested that the inclusion of calculated ocean metrics such as N^2 (related to vertical stability of the water column) and $O_{2,sat}$ (related to physical factors, biological production, and respiration) offered negligible improvements to the model because correlations

with Θ , S_A , p , and O_2 together encode much of the same information. Notably, the inclusion of seasonal variables sz_1 and sz_2 improved performance specifically in the upper 100 m (Figure S4c). An independent evaluation of our model framework using synthetic float data further supported the ability of RFR to reconstruct nitrate distributions as observed from a rapidly-sampling platform. When the synthetic RFR, trained using the selected model parameters on data from nine virtual floats, was applied to 4-hourly synthetic data that emulated Seaglider sampling, the error distributions had low median absolute error (MAE) values of $0.119 \mu\text{mol kg}^{-1}$ for synSG-1 and $0.138 \mu\text{mol kg}^{-1}$ for synSG-2 with an interquartile range of the absolute errors (IQR-AE) of 0.193 and $0.156 \mu\text{mol kg}^{-1}$, respectively (Figure 4d). For both synthetic gliders, over 98% of errors were within the float nitrate uncertainty of $0.5 \mu\text{mol kg}^{-1}$ (Maurer et al. 2021). The synthetic RFR errors are smaller in magnitude compared to those computed from real world data during validation (Figure 4b–c); this performance is consistent with the fact that the numerical earth system model, while offering data with high spatial resolution, only represents a portion of the full spectrum of small-scale variability that exists in the true ocean.

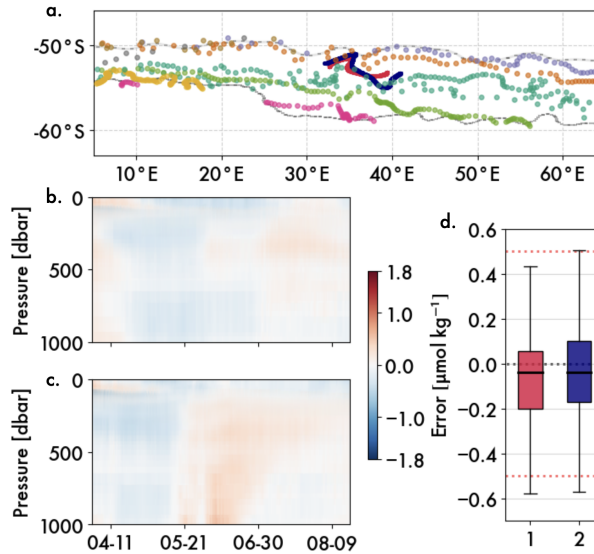


FIG. 4: Results of independent evaluation with synthetic floats. (a) Locations of synthetic float profiles at 10-day resampled resolution, and synthetic glider profiles at 4-hourly resolution for synSG-1 (red) and synSG-2 (navy). (b,c) Synthetic RFR nitrate estimation error (estimated – model truth; $\mu\text{mol kg}^{-1}$) for the upper 1000 dbar for synSG-1, synSG-2 respectively. Horizontal time axis in month-day format for 2019. (d) Boxplots of nitrate estimation errors ($\mu\text{mol kg}^{-1}$) from the synthetic gliders synSG-1 and synSG-2.

377 The RFR model trained on the nine BGC-Argo floats and GO-SHIP I06 with the final feature
 378 list, hereafter the SOGOS RFR, was tested on two different independent datasets (SOGOS float
 379 and GO-SHIP I07) to produce unbiased estimates of error within the spatiotemporal context of the
 380 Seaglider application. The SOGOS RFR returned errors largely under the float nitrate uncertainty
 381 with a test MAE (IQR-AE) of 0.208 (0.281) $\mu\text{mol kg}^{-1}$ when applied to the withheld SOGOS
 382 float data. When computed only over the Seaglider deployment (yeardays 120–200), the test MAE
 383 (IQR-AE) was slightly higher at 0.302 (0.343) $\mu\text{mol kg}^{-1}$; this difference may be due to the fact that
 384 the SOGOS deployment began in one of the most turbulent and therefore intensely variable regions
 385 of the global ocean. Even so, 87% of all test MAEs were $\leq 0.5 \mu\text{mol kg}^{-1}$ (BGC-Argo float nitrate
 386 measurement uncertainty; Maurer et al. 2021). The test errors exhibited significant variability both
 387 in time (horizontal axis) and as a function of pressure (vertical axis in Figure 5d). Below the MLD
 388 (magenta lines, Figure 5d), the RFR resulted in slight overestimation, whereas within the mixed
 389 layer there are different patterns of alternating over- and under-estimation that could reflect nitrate
 390 variability not adequately captured by the predictors used here. The error distribution is closely
 391 centered around 0 (Figure 5c) with a mean bias of $-0.0076 \mu\text{mol kg}^{-1}$, corresponding to a 0.08%
 392 relative error, and 95% of errors were between -0.704 and $0.689 \mu\text{mol kg}^{-1}$. The validation errors
 393 exhibited a larger spread in the upper ~ 300 m as expected given more intense variability in the
 394 upper ocean. When we applied the SOGOS RFR to GO-SHIP I07 data, we found comparable
 395 results to those seen with the SOGOS float (Figure S5). The test MAE (IQR-AE) of 0.257 (0.383)
 396 $\mu\text{mol kg}^{-1}$ corresponded to a 0.84% relative error and overall, 75% of errors were under the float
 397 nitrate uncertainty of $0.5 \mu\text{mol kg}^{-1}$ (Maurer et al. 2021).

398 We also compared SOGOS RFR errors from the SOGOS float to those from two existing machine
 399 learning algorithms that have been widely applied for biogeochemical estimation, CANYON-B
 400 (Bittig et al. 2018) and ESPER-Mixed (Carter et al. 2021). These algorithms were trained on
 401 ship-based data from different releases of the Global Data Analysis Project (Olsen et al. 2019).
 402 CARbonate system and Nutrients concentration from hYdrological properties and Oxygen using
 403 a Neural-network version B (CANYON-B) is the second iteration of a Bayesian neural network
 404 approach to predict nutrient variables (nitrate, phosphate, silicate) from a set of features including
 405 T, S, p, O_2 , latitude, longitude, and/or day of the year (Bittig et al. 2018). The Empirical Seawater
 406 Property Estimation Routines (ESPERs) are predictive algorithms (Carter et al. 2021) that can

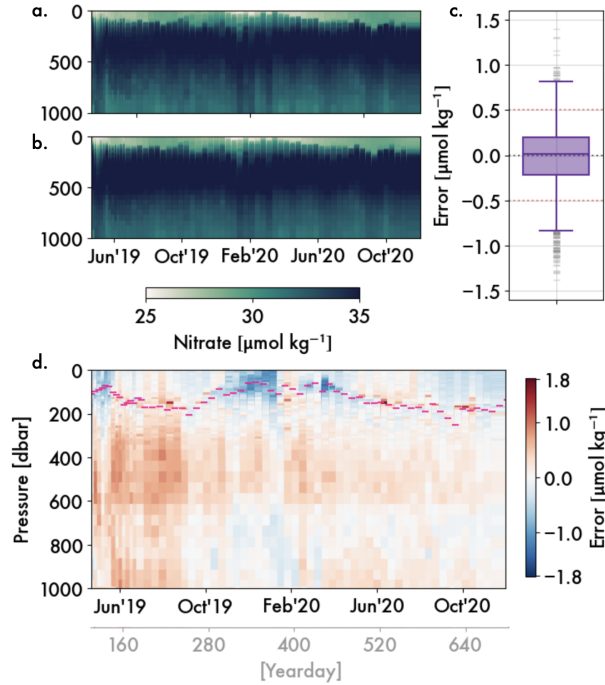


FIG. 5: Nitrate ($\mu\text{mol kg}^{-1}$) (a) observed by the SOGOS float and (b) estimated by the SOGOS RFR for the upper 1000 dbar. (c) Boxplot of nitrate errors ($\mu\text{mol kg}^{-1}$); outliers in gray dashes. (d) Nitrate error (estimated – observed; $\mu\text{mol kg}^{-1}$) over the upper 1000 dbar, MLD in magenta lines. Horizontal time axis in month 'year format; corresponding yearday relative to 2019 in gray.

estimate numerous biogeochemical properties, including nitrate, from a variety of feature lists; the updated ESPER-Mixed algorithm combines results from locally interpolated regressions (Carter et al. 2016) and neural networks. To estimate nitrate from ESPER-Mixed, we used input features of T, S, p, O_2 , latitude, and longitude; for CANYON-B, the same features plus time were considered.

Because the ship-based dataset used to train both CANYON-B and ESPER-Mixed has large observing gaps over the Southern Ocean, we expect that the addition of local BGC-Argo observations to our RFR training dataset will improve representation of relationships specific to the SOGOS region (e.g. those in Figure S8). The two ship-based algorithms performed similarly to each other, with excellent performance at depths greater than 200 m (Figure 6b-c), where nutrient concentrations vary at larger spatiotemporal scales than in the upper ocean. Their strong performance in the ocean interior underscores the utility of globally-trained algorithms for estimating large-scale nutrient distributions. However, both CANYON-B and ESPER-Mixed persistently and substantially underestimated nitrate in the mixed layer as compared to the locally-trained RFR (Figure 6a,d). The surface underestimation may be due to the fact that cruise observations used to train

421 CANYON-B and ESPER-Mixed are heavily biased towards austral summer (Bittig et al. 2018).
 422 Mixed layer nitrate is expected to be lower during these months when biological utilization is most
 423 active, in contrast to fall and winter when the mixed layer deepens and productivity decreases to
 424 low background levels (Su et al. 2022). CANYON-B, which considers day of year, returned better
 425 predictions in the upper 200 m than those given by ESPER-Mixed, which does not use time as a
 426 predictor. This difference mirrors the improvement found in the surface ocean for our RFR model
 427 upon addition of seasonal variables (Supplemental Material).

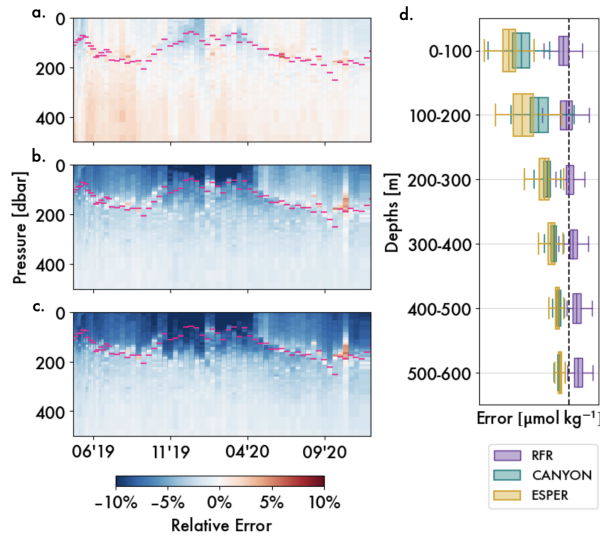


FIG. 6: Relative nitrate errors $[(estimated - observed) / observed]$ in the upper 500 dbar using (a) RFR, (b) CANYON-B, and (c) ESPER-Mixed using SOGOS float observations. Results not shown between 500 dbar and 1000 dbar are similar to those under ~ 300 dbar. MLD in magenta lines. Horizontal time axis in month 'year format. (d) SOGOS float errors ($\mu\text{mol kg}^{-1}$) in 100-m depth bins for RFR (purple), CANYON-B (teal), ESPER-Mixed (yellow).

428 Estimating time- and space-resolved nutrient distributions is especially challenging in the highly
 429 dynamic mixed layer. For both existing ship-based algorithms, previous evaluations noted that the
 430 seasonality and exact values of the mixed layer output were associated with significant uncertainties
 431 (Bittig et al. 2018; Carter et al. 2021). The reduced upper ocean biases from the SOGOS RFR
 432 support using additional local training data within the specific spatiotemporal bounds of the
 433 expected application. As such, the growing BGC-Argo observing array, which samples consistently
 434 throughout the seasonal cycle, can supplement training datasets for targeted regressions that better
 435 represent seasonal and spatial variability.

b. SOGOS Seaglider Nitrate Analyses

After model validation and testing, the two SOGOS Seaglider datasets were used as inputs to the SOGOS RFR to produce nitrate estimates along their tracks to ~ 1000 m depths (Figures 7b, 8c, and 8g; full sections in Figure S1). The average horizontal distance between Seaglider profiles was ~ 1.5 km (downcast and upcast are separate profiles), whereas float profiles were separated by ~ 70 – 80 km. Application of the SOGOS RFR therefore resulted in a ~ 50 -fold increase in horizontal resolution for the nitrate distributions (Figure 7b). Likewise for temporal resolution, the Seaglider profiles were separated (at mid-depth for their V-shaped dives) by ~ 2 – 3 hours while the SOGOS float returned profiles every 5 days, which was already twice as frequent as the 10-day sampling period of standard Argo floats. Overall, the SOGOS RFR generates two novel nitrate datasets in this turbulent subregion of the Southern Ocean by taking advantage of both regionally specific nutrient information captured by standard BGC-Argo floats and the high spatiotemporal resolution provided by Seagliders.

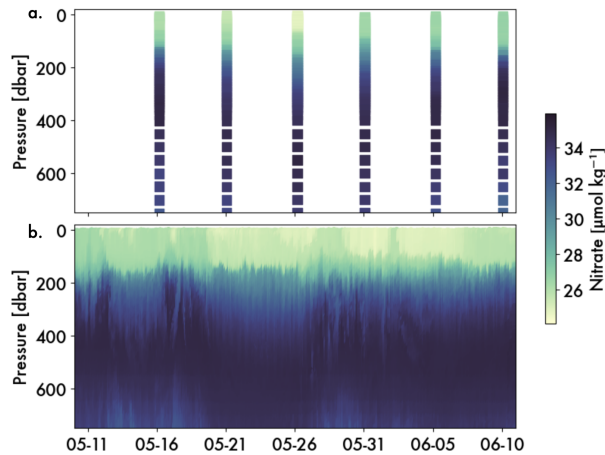


FIG. 7: Nitrate ($\mu\text{mol kg}^{-1}$) (a) observed by the SOGOS float over a period of ~ 30 days (6 profiles) and (b) predicted by the SOGOS RFR for Seaglider SG660 over the same time period (428 profiles). Horizontal time axis in month-day format for 2019.

1) MIXED LAYER TIME SERIES

Based on the SOGOS RFR depth-resolved distributions, small-scale variability in nitrate was most evident below the base of the mixed layer (Figure 8). Θ and S_A are impacted by atmospheric surface forcing (including heating, cooling, evaporation, and precipitation) that typically occurs

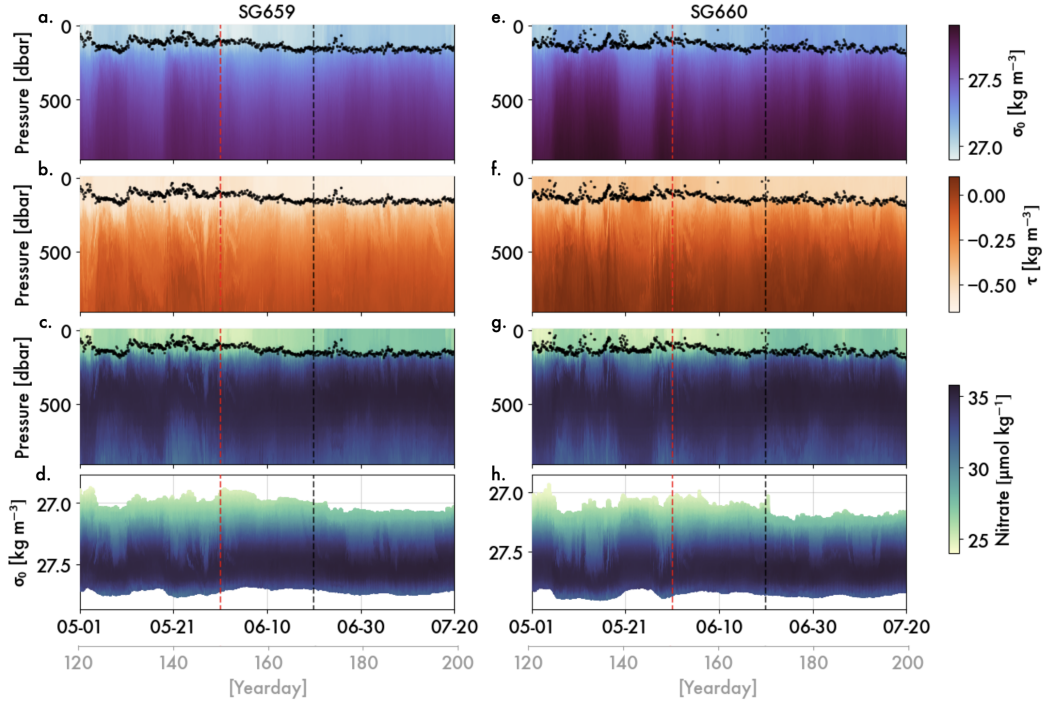


FIG. 8: (a) Potential density referenced to surface (σ_0 ; kg m^{-3}), (b) spice (τ ; kg m^{-3}), and (c) nitrate ($\mu\text{mol kg}^{-1}$) sections from SG659 plotted in time-pressure space; (d) nitrate plotted in time-density space. (d-f) Same variables as left, for SG660. Dashed vertical red and black lines indicate the end of the high EKE region (yeardays 120–150) and beginning of the low EKE region (yeardays 170–200), respectively. Horizontal time axis in month-day format; corresponding yearday of 2019 in gray. Full nitrate estimates also given in Figure S1.

at larger scales, such that small-scale variability generated by stirring tends to be dampened near the surface relative to the interior. The use of Θ and S_A as features in the SOGOS RFR could contribute to similar erasure of rapid variability in nitrate in the mixed layer. Outside of potential model artifacts, biological utilization of nitrate near the surface would also tend to decrease nitrate variability at short timescales (Lévy et al. 2018).

We identified connections between small-scale mixing and mixed layer nitrate variability by comparing two distinct oceanographic regimes sampled by the Seagliders: a high EKE region around the SWIR (yeardays 120–150) and a low EKE region downstream (yeardays 170–200; Figures 1a, 9a). When the Seagliders sampled the high EKE region (yeardays 120–150), they observed intermittent shoaling of the mixed layer (Figure 9b) as well as enhanced horizontal buoyancy gradients ($\nabla_h b$) that indicate submesoscale structures (Figure 9c). These findings aligned with characterization of the SOGOS region by Dove et al. 2021, which demonstrated that the high

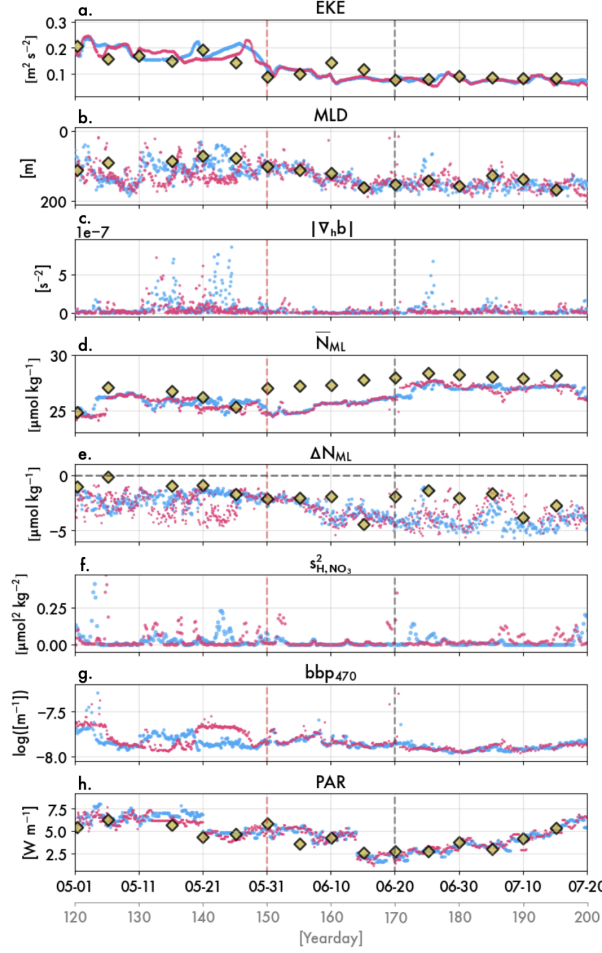


FIG. 9: For SG659 (teal), SG660 (magenta) and SOGOS float (yellow diamonds, when available), (a) along-track eddy kinetic energy (EKE); (b) mixed layer depth (MLD); (c) horizontal buoyancy gradient ($\nabla_h b$) in the mixed layer; (d) mixed layer mean nitrate concentration ($\bar{N}_{ML}$; 95% C.I. $\pm 0.042 \mu\text{mol kg}^{-1}$); (e) difference in nitrate concentration across the base of the mixed layer (ΔN_{ML} ; 95% C.I. $\pm 0.390 \mu\text{mol kg}^{-1}$); (f) horizontal variance in mean mixed layer nitrate (s^2_{H,NO_3}) for 10-profile window (~ 2 days); (g) logarithm of mixed layer mean backscatter at 470 nm (bbp_{470}), (h) photosynthetically active radiation (PAR). Horizontal time axis in day-month format; corresponding yearday of 2019 in gray. In each panel, the dashed red and black lines indicate the end of the high EKE region (yeardays 120–150) and beginning of the low EKE region (yeardays 170–200), respectively. Panels d–f use the RFR-predicted nitrate fields while the rest are observed quantities.

EKE region is rich in submesoscale instabilities that affect vertical stratification at the base of the mixed layer and promote greater biogeochemical exchange with the ocean interior.

The mean mixed layer nitrate $\bar{N}_{ML}$ from the SOGOS RFR was between 24.13 and $27.85 \mu\text{mol kg}^{-1}$ (95% C.I. $\pm 0.042 \mu\text{mol kg}^{-1}$), with values in the high EKE region reduced compared to those in the low EKE region (Figure 9d). Significant differences in $\bar{N}_{ML}$ observed by the platforms were

occasionally found (e.g. yeardays 150–165; Figure 9d), which were partially due to increased spatial separation (Figure 1b). Changes in $\bar{N}_{ML}$ could reflect mixing between surface waters and deeper, nutrient-rich waters, either through deepening of the mixed layer itself or through exchange along sloping isopycnals that intersect the base of the mixed layer (Freilich and Mahadevan 2019). The injection of nutrients into the mixed layer could also spur primary productivity by relieving nutrient limitation (primarily caused by low iron levels in the surface Southern Ocean), acting to reduce $\bar{N}_{ML}$. The lower $\bar{N}_{ML}$ values in the high EKE region coincided with increased optical backscatter (bbp_{470} ; Figure 9g), which indicated elevated concentrations of particulate organic carbon in the mixed layer. Together with satellite-based light availability (PAR; Figure 9h), these time series provided evidence of increased biological productivity in the high EKE region. Because these data were collected in austral fall when nutrients were depleted by summer blooms and light availability was decreasing, background biological activity was expected to be relatively low (Su et al. 2022). However, additional nutrient input into the mixed layer could still spur productivity, even if full bloom initiation is delayed or occurs only weakly.

As with MLD and $\nabla_h b$, the Seagliders observed rapid changes in upper ocean nitrate that are not captured by the coarse sampling of the SOGOS float. Within the high EKE region (yeardays 120–150), the difference in nitrate across the base of the mixed layer (ΔN_{ML}) observed by the SOGOS float was consistently small in magnitude (-1.69 to -0.15 $\mu\text{mol kg}^{-1}$). In contrast, the RFR Seaglider ΔN_{ML} changed rapidly and covered a much larger range during the same period (-4.72 to -0.88 $\mu\text{mol kg}^{-1}$; 95% C.I. $\pm 0.39 \mu\text{mol kg}^{-1}$; Figure 9e). Furthermore, the SOGOS RFR estimates revealed elevated values of horizontal variance in mixed layer nitrate (s_{H,NO_3}^2) in the high EKE region of enhanced stirring (Figure 9f). The timing of elevated s_{H,NO_3}^2 coincides with sharp MLD gradients and $\nabla_h b$ observed by the Seagliders (e.g. yeardays 125, 131, 137, 148 for SG660). The SOGOS RFR estimates thus appear to resolve filaments of water masses with distinct nutrient characteristics, sourced from different regions, as they are stirred by motions at the meso- and submesoscales. Overall, this analysis provides evidence of enhanced submesoscale nutrient upwelling through the steepening of density gradients and weakening of stratification at the base of the mixed layer.

498 2) TIMESCALES OF NUTRIENT VARIABILITY

499 Using the RFR nitrate distributions, we assessed interior ocean nitrate variability across a range
500 of timescales, corresponding to periods of ~ 5 hours to ~ 50 days. Nitrate and spice were tracked on
501 four density surfaces (“along-isopycnal” signals) with average corresponding depths ($\bar{d}$) ranging
502 from ~ 170 – 400 m for SG660 (Table 1; Figure 8h). We applied continuous wavelet transform
503 (CWT) to determine when small-scale upper ocean variability was enhanced, with higher CWT
504 amplitudes indicating stronger variability.

σ_0 (kg m $^{-3}$)	$\bar{d}$ (m)
27.30	174.63
27.40	223.68
27.50	296.80
27.60	396.92

TABLE 1: Average depth ($\bar{d}$) of analyzing isopycnal (σ_0) for SG660

505 The along-isopycnal nitrate CWTs reflected frequencies of nitrate variability in both the
506 mesoscale and submesoscale range. At mesoscale periods ≥ 20 days, there were significant
507 bands of high CWT amplitudes in both nitrate and spice when the Seaglider sampled the eddy-rich,
508 high EKE region (yeardays 140–150; Figure 10a–b). This strong mesoscale variability in the
509 along-isopycnal signals extended throughout the upper ~ 400 m (Figure 10). At shorter periods
510 (~ 0.2 days to ~ 2 days), there were transient bursts of enhanced submesoscale variability in both
511 nitrate and spice when the platforms sampled the high EKE region (e.g. yeardays 130, 137, 150).
512 The short-lived events tended to occur where there were sharp gradients in mixed layer depth and
513 shoaling of the analyzing isopycnal (e.g. yeardays 130, 137), presumably due to eddy activity
514 (Figure 11a–c). At these times, the Seaglider appeared to sample across filaments of enhanced
515 FSLE (Figure 11d–f), which are typically found around submesoscale fronts or between mesoscale
516 eddies (Siegelman et al. 2020). However, some observed large changes in MLD were not associ-
517 ated with enhanced submesoscale variability in the nitrate CWTs (e.g. yeardays 124, 147, Figure
518 11a–c). Small bursts of weaker submesoscale variability continued to occur sporadically in the
519 low EKE region along the shallow isopycnals $\bar{d} \leq 296$ m (Figure 10).

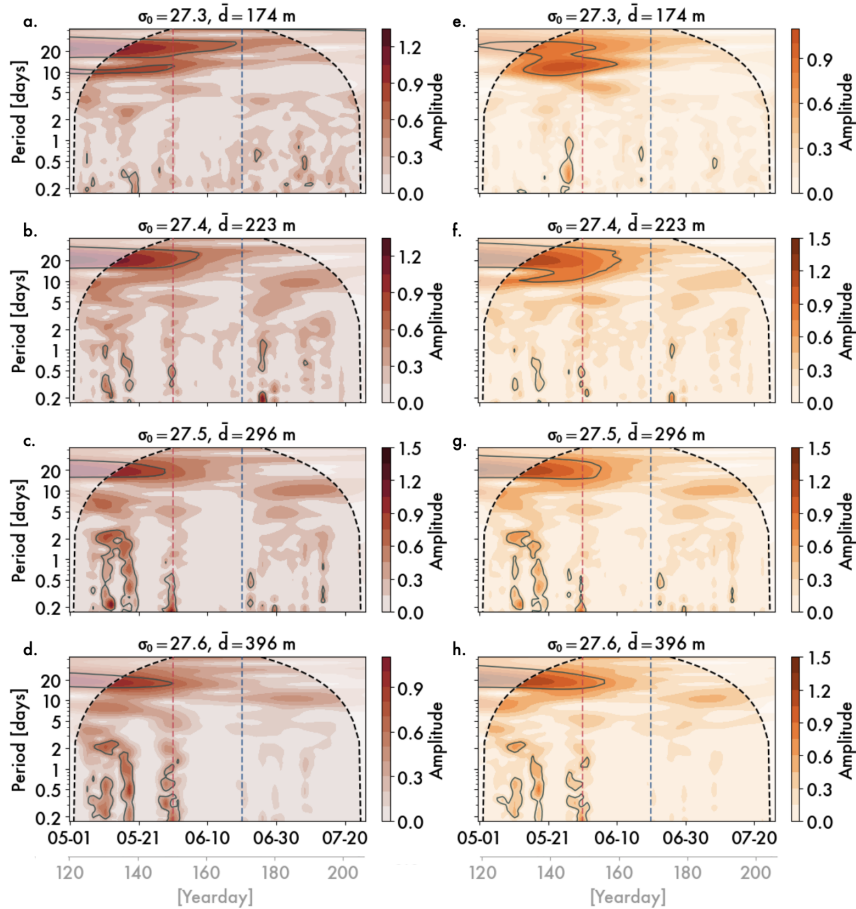


FIG. 10: (a-d) CWTs of SOGOS RFR nitrate estimates from SG660 along four analyzing isopycnals (1). (e-h) CWTs of observed spice from SG660 along the same isopycnals; 95% significance contours in dark gray. In each panel, the dashed vertical red and blue lines indicate the end of the high EKE region (yeardays 120–150) and beginning of the low EKE region (yeardays 170–200), respectively. Inertial period at this latitude is ~ 0.643 days, or 15.4 hours. Results are significant only within the cone of influence (COI) in dashed black, which reflects the finite nature of the time series (Torrence and Compo 1998); results insignificant in white shaded region.

Overall, nitrate variability in the SOGOS RFR distributions was often dominated by mesoscale modulation of water masses at periods of ~ 20 – 50 days. However, especially in the high EKE region, rapid changes in along-isopycnal nitrate occurred during transient events at timescales associated with submesoscale stirring (~ 0.5 to 2 days). Although not within the scope of this paper, wavelet transform coherence could be used to explore correlations between physical and biogeochemical CWTs (Foster 1996). Continued progress in estimating high-resolution nutrient distributions, here achieved by RFR, will invite new methods of quantifying biogeochemical variability.

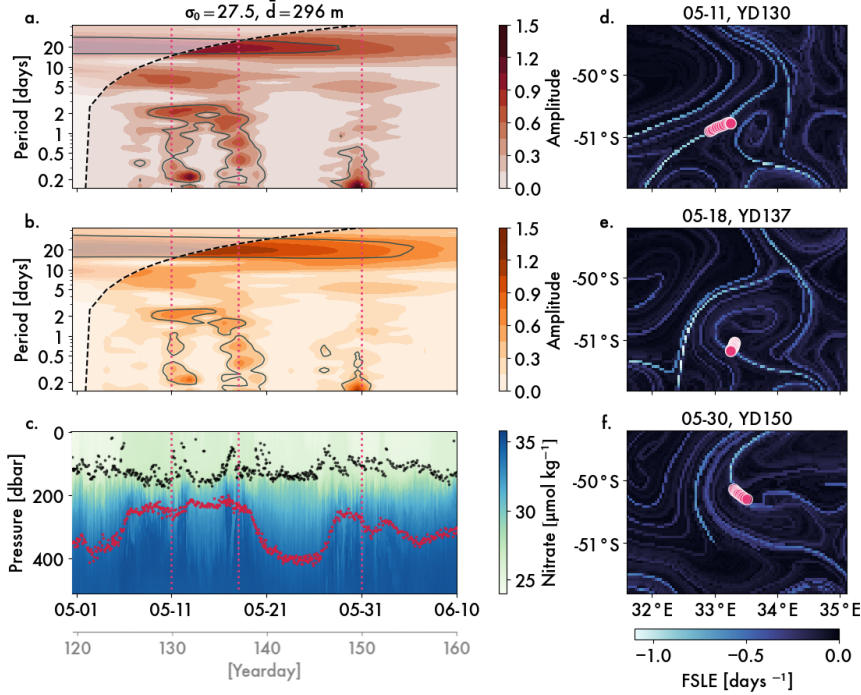


FIG. 11: (a, b) Nitrate and spice CWTs from SG660 for analyzing isopycnal $\sigma_0 = 27.5$ ($\bar{d}=296$ m); 95% significance contours in dark gray. COI in dashed black; results insignificant in white shaded region. (c) Seaglider SG660 nitrate section with analyzing isopycnal $\sigma_0=27.4$ in red dots; MLD in black dots. Horizontal time axis in month-day format; corresponding yearday of 2019 in gray. Dotted magenta lines in panels a–c indicate yeardays represented in panels d–f. (d–f) Surface FSLE at daily resolution, with SG660 profile locations in magenta for a given yearday.

5. Discussion and Conclusions

In this study, we used random forest regression (RFR) to infer information on rapid nitrate variability in the energetic Southern Ocean using observations from the SOGOS field experiment. Seagliders provided measurements of ocean properties at high resolution, while a co-deployed BGC-Argo float provided an optimal dataset for testing the RFR within the specific spatiotemporal context of the SOGOS deployment. The locally trained RFR used nearby BGC-Argo observations alongside shipboard measurements to capture region-specific nitrate variability and returned lower test errors than those obtained with existing ship-based routines. Our approach highlights that sustained sampling of the Southern Ocean by autonomous floats fills critical gaps in traditional observational coverage, especially over winter months when the Southern Ocean is largely inaccessible to ships (Sarmiento et al. 2023).

By applying the SOGOS RFR to generate novel nitrate distributions along the rapid-sampling Seaglider tracks, we observed that vigorous submesoscale stirring downstream of the Southwest Indian Ridge was associated with sharp changes in mixed layer nitrate. Quantifying the dominant timescales of variability with wavelet analysis suggested that nutrients were sporadically injected into the upper ocean in small-scale filamentary structures. Despite significant periods of enhanced variability at submesoscales, these were only evident in the RFR Seaglider estimates and not in the BGC-Argo float observations. Likewise, although the coarser-resolution float observations were unable to distinguish nitrate characteristics between the high and low EKE regions, the SOGOS RFR Seaglider estimates better resolved the distinct distributions (Figure 12) and returned a statistically significant difference between $\bar{N}_{ML}$ in the two regions (Welch t-test statistic: -49.6, p-value ≈ 0 , dof = 1349). Our RFR approach thus extended the benefits of rapid sampling to variables not directly represented in the Seagliders' sensor array. We encourage future deployments to consider utilizing heterogeneous arrays of instruments, especially where machine learning can be applied to fill in missing information (e.g. Salam and Hsieh 2023; Salcedo-Sanz et al. 2020; Renosh et al. 2023; Lermusiaux et al. 2017; Chai et al. 2020).

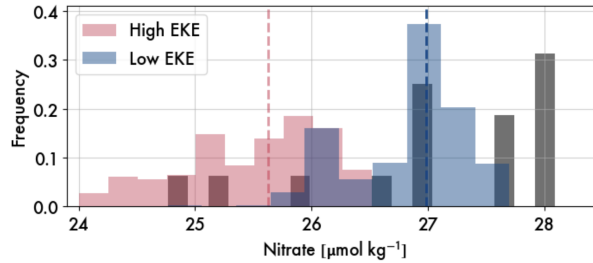


FIG. 12: Distributions of $\bar{N}_{ML}$ from SOGOS RFR estimates in the high (red) versus low (blue) EKE regions; solid lines indicate distribution means. Float observations shown in gray.

Among machine learning approaches, RFR is a relatively simple algorithm that can be trained on regional datasets smaller than those required for deep learning. Simple learners like RFR can offer greater stability and interpretability compared to deep learning methods, with equivalent performance given limited data (Domingos 2012). RFR has been successfully used for both physical and biogeochemical ocean analyses, including estimation of surface carbon (Gregor et al. 2019; Sharp et al. 2022), oxygen (Sharp et al. 2023; Giglio et al. 2018), heat content anomalies (Lyman and Johnson 2023), ocean color (Chen et al. 2019), surface waves (Callens et al. 2020), and eddy diffusivity (Zhang et al. 2023). RFR is especially attractive for geospatial applications

because of its reduced overfitting tendency and ability to capture nonlinear relationships (Stock 2022).

Given that ocean relationships vary strongly over space and time, our RFR does not aim to capture a nitrate mapping that is fully generalizable over the global ocean (Millard and Richardson 2015). Since geospatial regression models can struggle to extrapolate outside the bounds of training data, the choice of training observations should be tailored for a given application (Heimdal et al. 2024; Millard and Richardson 2015). In our case, measurements from nine BGC-Argo floats (Figure 2) provided sufficient local observations for the RFR to outperform algorithms trained on ship-based datasets with sparse coverage of the Southern Ocean, especially for upper ocean estimation. Information from these nine floats is likely still insufficient to fully resolve the dynamic relationships among all ocean properties analyzed here. Furthermore, our model relied on the assumption that nitrate concentration correlates significantly with included predictive variables, such as Θ , S_A , and O_2 , but these correlations may be weaker in different parts of the Southern Ocean (Ishizu and Richards 2013). In fact, the same submesoscale processes we aim to diagnose using the high-resolution RFR estimates may lead to greater decoupling between nitrate and other variables (Mahadevan 2016; Omand and Mahadevan 2013). Despite these limitations, our locally-trained RFR produced remarkably low errors in one of the most turbulent areas of the global ocean and demonstrated the potential for machine learning to improve mapping of oceanic fields.

Machine learning and artificial intelligence are being increasingly applied for oceanographic studies (Sonnewald et al. 2021; Sun et al. 2022), including improving satellite altimetry products (Martin et al. 2023; Fan et al. 2021), estimating biogeochemical distributions (Bittig et al. 2018; Carter et al. 2021), and identifying eddy activity (Ashkezari et al. 2016; Zhang et al. 2023), among many others. Future application of RFR or similar machine learning approaches can be used to synthesize information from diverse, multi-platform oceanographic datasets (Salam and Hsieh 2023; Salcedo-Sanz et al. 2020). Here, a RFR model was developed to reconstruct upper ocean nitrate information during a 2019 Southern Ocean field campaign at much higher resolution than was directly observed, which provided valuable observation-based insights into small-scale ocean biogeochemical variability. A growing record of global observations from Argo floats and other autonomous instruments presents exciting opportunities for locally-trained regressions to derive additional value from targeted observing experiments with sometimes incomplete datasets. Such

591 efforts to bridge data gaps using new ocean technologies and machine learning techniques will
592 expand our knowledge of global biogeochemical cycles at previously inaccessible scales.

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Data availability statement. The SOGOS Seaglider data are archived at NOAA’s National Centers for Environmental information and can be accessed at <https://www.ncei.noaa.gov/archive/accession/0228185> and <https://www.ncei.noaa.gov/archive/accession/0228187>. Processed glider datasets are also available through Balwada 2023 (DOI 10.5281/zenodo.8361656), and the SOGOS RFR nitrate estimates through Song 2025 (DOI 10.5281/zenodo.14510704). Argo float data were collected and made freely available by the International Argo Program and the national programs that contribute to it (<https://argo.ucsd.edu>, <https://www.ocean-ops.org>). The Argo Program is part of the Global Ocean Observing System; Argo float data and metadata are from Global Data Assembly Centre (Argo 2021; DOI 10.17882/42182). Shipboard data were collected and made publicly available by the International Global Ship-based Hydrographic Investigations Program (GO-SHIP; <http://www.go-ship.org/>) and the national programs that contribute to it. The satellite altimetry data are freely available through the E.U. Copernicus Marine Environment Monitoring Service (CMEMS; DOI 10.48670/moi-00148), and the value-added FSLE product is provided by Archiving, Validation and Interpretation of Satellite Oceanographic data (LOCEAN/CLS/CTOH/CNES 2021; DOI 10.24400/527896/a01-2022.002). MODIS-Aqua satellite

data are hosted by NOAA and provided through the NASA Ocean Biology Processing Group (<https://coastwatch.pfeg.noaa.gov/erddap/griddap/erdMH1par08day.html>).

APPENDIX

Data Quality Control and Processing

a. Seaglider Processing

Seaglider data was reprocessed into L2 xarray Datasets (courtesy of Geoff Shilling and Craig Lee, Applied Physics Lab) which organize measurements into profiles and average raw observations in 1 m depth intervals to 1000 m. We used the despiked L3 product, which interpolates observations vertically (gaps ≤ 50 m in depth) and removes outliers >2 standard deviations from the running mean. Additive corrections for Θ and S_A were determined by matching Seaglider data to the closest quality-controlled BGC-Argo and ship measurements, using a difference threshold of ~ 5 m depth and ~ 10 km distance consistent with float and bottle match thresholds for standard BGC-Argo quality control (Maurer et al. 2021). The Seaglider offsets (dT and dS; Seaglider minus float) had statistically insignificant slopes along depth, so a single additive correction was made for each Seaglider. For temperature, 0.0629°C was added to all profiles from SG659, and 0.030°C for SG660. For salinity, SG659 had negligible corrections while 0.18 psu was subtracted from SG660 measurements (comparable to those performed in Dove et al. 2021).

Seaglider oxygen optode data were corrected for sensor time lag, which typically results in slight overestimation of oxygen during descent and slight underestimation during ascent, using adapted methods for oxygen corrections on Argo floats (code courtesy of Yuichiro Takeshita, Stoer et al. 2023). For the standard foil Aanderaa optodes, we used a boundary layer thickness of ~ 40 μm and time response calculated internally following Bittig et al. 2018. The oxygen data were corrected using a gain correction (Johnson et al. 2015) as $O_{2corr} = 1.126(O_2) - 3.256$ for SG659, and $O_{2corr} = 1.0866(O_2) - 0.146$ for SG660.

b. GO-SHIP Processing

Bottle data from the GO-SHIP cruises I06 and I07 in 2019 were accessed through the CLIVAR and Carbon Hydrographic Data Office (CCHDO; <https://cchdo.ucsd.edu/cruise/325020190403>; <https://cchdo.ucsd.edu/cruise/49NZ20191229>). Quality control was performed using the provided flags (2: no problems noted).

c. BGC-Argo Processing

Delayed-mode BGC-Argo data from ten floats (WMO IDs: 5904469, 5904659, 5905135, 5905368, 5905996, 5906007, 5906030, 5906031, 5906207, 5906211) were downloaded from an Argo Global Data Assembly Center (GDAC) using the Python BGC-Argo Toolbox (<https://github.com/go-bgc/workshop-python>). Quality control was performed using standard BGC-Argo QC flags (1: good data, 2: probably good data, 8: interpolated value).

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